

ONLINE SHOPPING BEHAVIOUR ANALYSING CUSTOMER ACTIVITY AND PREDICTING PURCHASE INTENT

Project Report Submitted in Fulfillment of the Requirements for the IDRA Data Science and AI
Training Program

Prepared by: SARATHI R

Institute: SRM UNIVERSITY AP

Institute Roll No.: AP25122230040

Enrollment No.: IDRA-2026-889462

Under the Supervision of Dr. Shaheena Salam

India Data Research Academy

September 2026

DECLARATION

I declare that this report presents my own analysis of the supplied e-commerce dataset. The code, analysis, figures and interpretations were prepared for the IDRA Data Science and AI capstone. Sources used for dataset context and software documentation are acknowledged in the references.

ACKNOWLEDGEMENT

I acknowledge the India Data Research Academy, the Data Science and AI Training Program, Dr. Shaheena Salam, and my institute for the learning resources and guidance that supported this capstone.

ABSTRACT

This capstone examines 25,000 online shopping sessions to identify session-context and browsing signals associated with purchase intent. A final methodology audit removed `added_to_cart`, `payment_method` and `rating` from the predictive feature set because the supplied file does not demonstrate that they are available before the prediction point. The final model uses only leakage-safe session context, product, promotion, browsing and calendar fields plus two transparent engineered features. The selected Logistic Regression model achieved 53.8% accuracy, 26.5% precision, 59.7% recall, 36.7% F1-score and 0.567 ROC-AUC on the held-out test set.

Keywords: online shopping; purchase intent; classification; leakage audit; random forest; e-commerce analytics

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1. INTRODUCTION

Online retailers make decisions before a customer completes a purchase: which sessions deserve timely support, which campaigns attract high-intent visitors and where checkout friction may exist. This study estimates purchase intent from fields that are defensibly available before purchase completion.

2. PROBLEM STATEMENT AND MOTIVATION

Can purchase intent be predicted without using fields that reveal or follow the purchase outcome? The final model prioritizes methodological validity over inflated scores.

3. OBJECTIVES

- Audit the supplied data for completeness, duplicates and feature suitability.
- Perform a leakage audit on disputed fields before modeling.
- Construct a leakage-safe feature set for purchase-intent prediction.
- Compare classification models on a held-out test set.
- Translate the final results into cautious business recommendations.

4. RESEARCH QUESTIONS

- What is the prevalence of purchase sessions?
- Which supplied fields are safe to use before purchase completion?
- How accurately can leakage-safe session fields classify purchase intent?
- Which final model features are most useful for discrimination?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

The dataset was supplied for Project 10 as P_10_Ecommerce.csv. It contains one row per online shopping session. The target is purchased, where 1 indicates a completed purchase and 0 indicates no completed purchase.

Table 1. Dataset summary

Measure	Value
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Rows	25,000
Raw columns	29
Target	purchased (binary)
Purchase sessions	5,616
Purchase rate	22.5%
Time coverage	01 Jan 2024 to 30 Dec 2024

Table 2. Final feature groups

Group	Final variables	Role
Session context	device_type, user_type, marketing_channel	Categorical features
Product and promotion	product_category, unit_price, discount_percent, discounted_unit_price	Product and price features
Browsing engagement	pages_viewed, time_on_site_sec, pages_per_minute	Pre-outcome engagement features
Calendar	visit_month, visit_weekday, visit_season, session_duration_bucket	Categorical context
Target	purchased	Prediction target

6. DATA CLEANING

The raw audit found 0 missing cells and 0 exact duplicate rows. Date text was converted to a date type. No rows were removed.

7. DATA PREPROCESSING AND LEAKAGE AUDIT

The data were split into 20,000 training and 5,000 testing sessions using stratification and random_state=42. Numerical variables were median-imputed and standardized inside the pipeline; categorical variables were mode-imputed and one-hot encoded. Transformations were fitted only on the training set.

The final methodology audit found that added_to_cart, payment_method and rating are not demonstrably available before the prediction point in the supplied file. These variables were removed rather than justified by assumptions. The audit also retained the previous exclusion of revenue, revenue_normalized, cart_abandoned, review fields and identifiers.

Table 3. Leakage audit decisions

Field	Final decision	Reason
added_to_cart	Excluded	No event timestamp proves pre-prediction availability; cart path combinations encode the outcome.
payment_method	Excluded	Usually captured at payment/checkout; supplied file does not prove earlier availability.
rating	Excluded	Usually post-experience or review data; supplied file does not prove

		earlier availability.
revenue fields	Excluded	Outcome values (revenue and revenue_normalized).
cart_abandoned	Excluded	Reveals a non-purchase cart outcome.

8. EXPLORATORY DATA ANALYSIS

Purchase sessions account for 22.5% of the data. Figures compare outcome prevalence, coded marketing channel conversion and browsing engagement. Coded channels are treated as categories, not ranks.

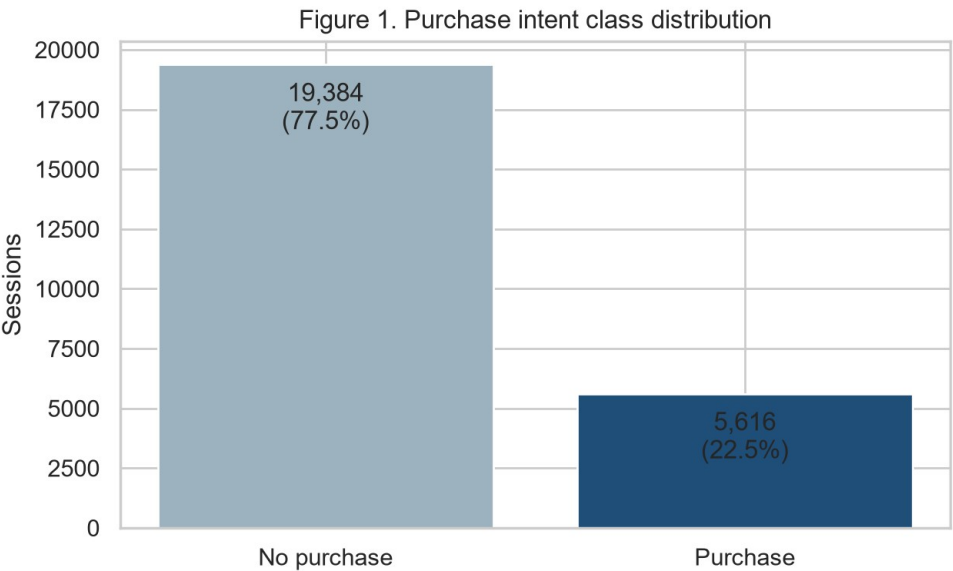


Figure 1. Purchase intent class distribution

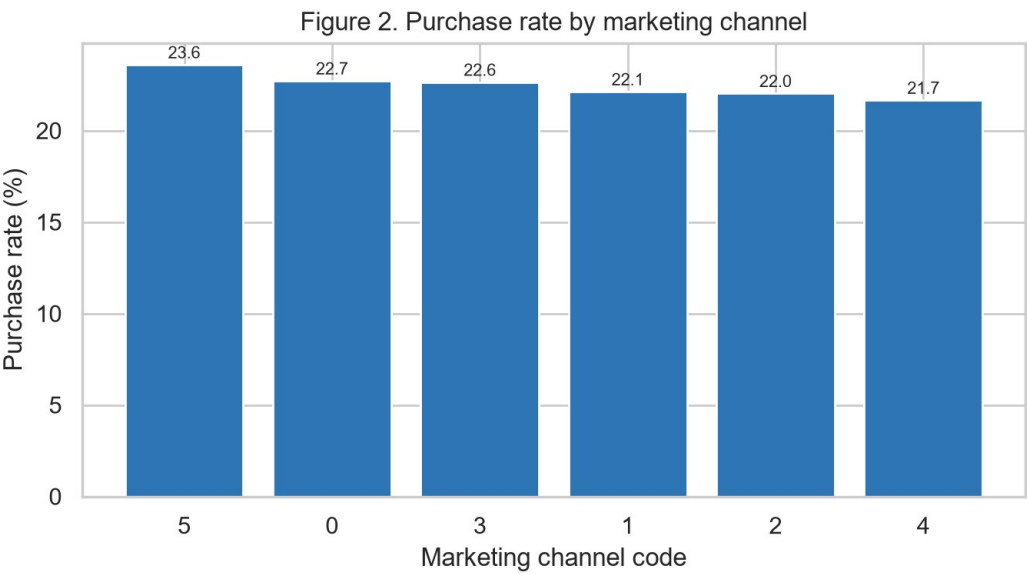


Figure 2. Purchase rate by marketing channel

Figure 3. Browsing engagement by purchase outcome

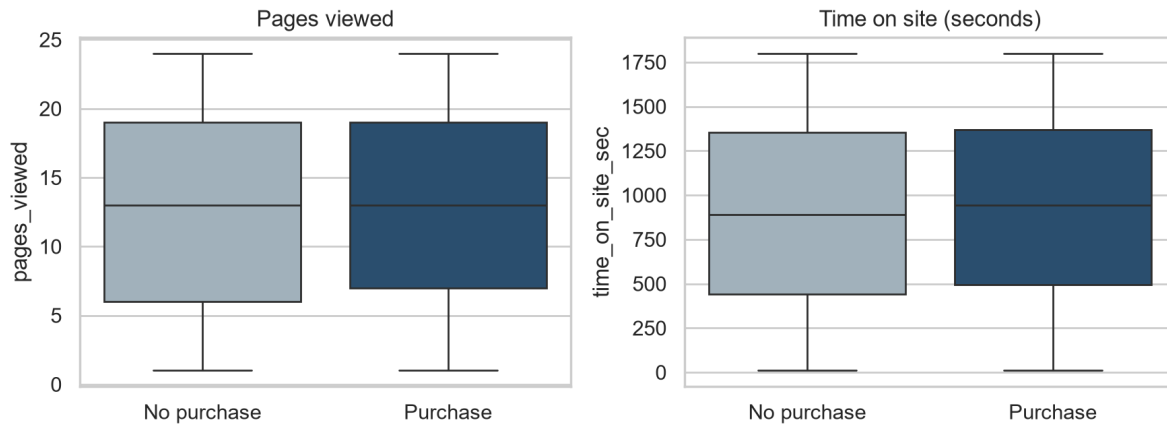


Figure 3. Browsing engagement by purchase outcome

9. STATISTICAL ANALYSIS

Descriptive comparisons were used rather than causal tests. Medians are reported for skew-sensitive browsing measures, and class proportions are used for the binary outcome.

Table 4. Median browsing engagement by outcome

Measure	No purchase	Purchase
Pages viewed	13.0	13.0
Time on site sec	890.0	942.5

Interpretation: purchase sessions have the same median page count (13.0) as non-purchase sessions, while median time on site is higher for purchase sessions (942.5 seconds versus 890.0 seconds). These are descriptive associations and should not be interpreted as causal effects.

10. FEATURE ENGINEERING

Two transparent features were derived from retained fields: `pages_per_minute` measures browsing pace, and `discounted_unit_price` applies `discount_percent` to `unit_price`. No target-derived feature was created.

Table 5. Final model feature list

Type	Features
Categorical	device_type, user_type, marketing_channel, product_category, visit_weekday, visit_month, visit_season, session_duration_bucket
Numeric	unit_price, discount_percent, pages_viewed, time_on_site_sec, pages_per_minute, discounted_unit_price

11. MODEL DEVELOPMENT

This is a supervised binary classification problem. Logistic Regression provides a linear baseline, and Random Forest captures non-linear relationships and feature interactions. The selected model is the one with the higher held-out F1-score.

12. MODEL EVALUATION

Overfitting assessment: Logistic Regression showed a very small training–test accuracy gap (53.2% training versus 53.8% test), so its accuracy does not show a substantial generalisation gap. Random Forest reached 99.97% training accuracy but 72.9% test accuracy, indicating a substantial training–test gap and a risk of overfitting. For interpretation, the held-out test results are therefore prioritised over training performance.

The selected model was Logistic Regression. Training accuracy was 53.2% versus test accuracy of 53.8%. The held-out test metrics are the relevant estimate for this report.

Table 6. Test-set model performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	53.8%	26.5%	59.7%	36.7%	0.567
Random Forest	72.9%	26.6%	11.8%	16.3%	0.532

The selected-model confusion matrix contains 2,022 true negatives, 1,855 false positives, 453 false negatives and 670 true positives.

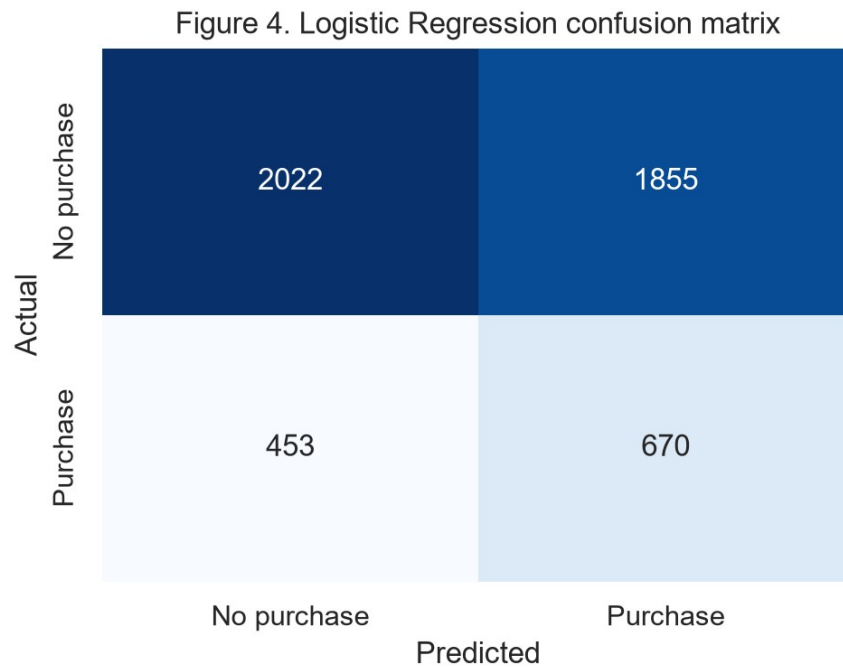


Figure 4. Logistic Regression confusion matrix

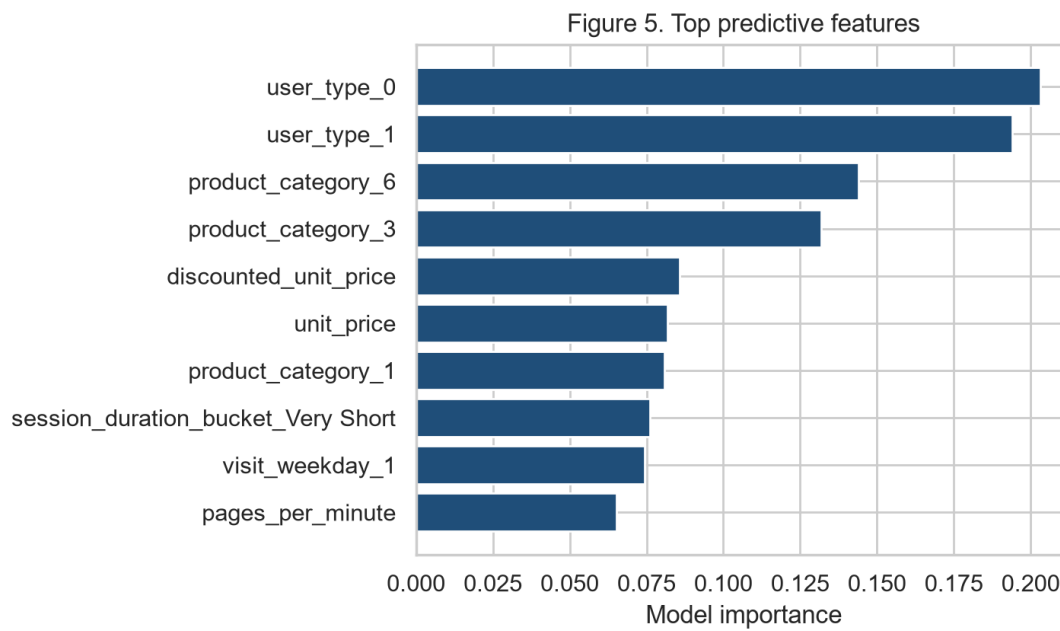


Figure 5. Top predictive features in the selected model

13. RESULTS AND FINDINGS

- The final leakage-safe Logistic Regression model achieved 36.7% F1-score and 0.567 ROC-AUC.

- Final performance is lower than the earlier leakage-affected run, which is expected after removing disputed outcome-linked fields.
- Browsing duration, browsing pace, price/promotion fields and categorical session context are the remaining evidence sources.
- The model should be treated as a prioritization aid, not a causal explanation.

14. DISCUSSION

The final results favor methodological defensibility over inflated accuracy. Removing `added_to_cart`, `payment_method` and `rating` reduces direct information about checkout or post-purchase behavior, leaving a cleaner estimate of how much can be learned from session context and browsing signals alone.

15. LIMITATIONS

- The dataset documentation does not provide event timestamps for all fields.
- The analysis is observational and does not demonstrate causal effects.
- A random split estimates same-source performance; time-based validation would be stronger.
- Business costs were not supplied, so the probability threshold was not optimized.

16. CONCLUSION

This capstone now presents a leakage-audited purchase-intent workflow. The selected Logistic Regression model achieved 53.8% accuracy, 26.5% precision, 59.7% recall, 36.7% F1-score and 0.567 ROC-AUC using only the final leakage-safe feature set.

17. RECOMMENDATIONS AND FUTURE WORK

- Collect timestamped event data so future models can define the prediction point precisely.
- Validate the model on later sessions before operational use.
- Set the classification threshold using business costs for false positives and false negatives.
- Add pre-session customer-history features only if they are known before the prediction point.

18. REFERENCES

India Data Research Academy. (2026). Capstone Project Guide 2026. Supplied course material.

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Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.

APPENDIX A. CODE AND REPRODUCIBILITY

The accompanying notebook contains the complete executable workflow: loading, cleaning, leakage audit, feature engineering, preprocessing, model comparison, evaluation, feature importance and output generation. The submission folder includes the notebook, cleaned dataset and final machine-readable analysis summary.